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Shape-optimization of round-to-slot holes for improving film cooling effectiveness on a flat surface

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Abstract

Single-objective optimization for improving adiabatic film cooling effectiveness is performed for single row of round-to-slot film cooling holes on a flat surface by using CFD analysis and surrogate approximation methods. Among the main geometric parameters, dimensionless hole-to-hole pitch (P/d) and slot length-to-diameter (l/d) are fixed as 2.4 and 2 respectively, and the other parameters (hole height-to-diameter ratio, slot width-to-diameter and inclination angle) are chosen as the design variables. Given a wide range of possible geometric variables, the geometric optimization of round-to-slot holes is carried out under two typical blowing ratios of $M = 0.5$ and $M = 1.5$ by selecting a spatially-averaged adiabatic film cooling effectiveness between $x/d = 2$ and $x/d = 12$ as the objective function to be maximized. Radial basis function neural network is applied for constructing the surrogate model and then the optimal design point is searched by a genetic algorithm. It is revealed that the optimal round-to-slot hole is of converging feature under a low blowing ratio but of diffusing feature under a high blowing ratio. Further, the influence principle of optimal round-to-slot geometry on film cooling performance is illustrated according to the detailed flow and thermal behaviors.

Nomenclature

d	film hole diameter (mm)
F	fitness function
I	momentum flux ratio
L	film hole length (mm)
l	slot length (mm)
M	blowing ratio
P	hole-to-hole pitch (mm)
s	slot width (mm)
T	temperature (K)
t	film hole height (mm)
u	velocity (m/s)
x	streamwise-direction
y	normal-direction
z	lateral-direction

Greek letters

ρ	density (kg/m^3)
θ	inclination angle of film hole (°)
η	film cooling effectiveness
Θ	dimensionless temperature
ω	vorticity (1/s)

Subscripts

ad	adiabatic
av	area-averaged
avx	laterally-averaged
aw	adiabatic wall
c	coolant or secondary flow
∞	primary flow

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1 Introduction

With the advancement of gas turbine power technology, the gas temperature approaching the turbine inlet is continuously increased. To meet the increasingly rigorous demand of protecting the hot section components from overheat, it is an urgent mission for thermal engineers to improve the cooling efficiency. As film cooling is an indispensable means to the high efficient cooling configuration, the improvement of film

cooling effectiveness is meaningful for improving the cooling efficiency, which is still a remarkable issue by now.

According to the previous investigations, the film cooling from two-dimensional continuous slots is confirmed to be an ideal cooling scheme for achieving the best film cooling effectiveness. However, due to many really competing constraints in the practical application, two-dimensional continuous slots are generally impractical to be directly adopted in the high-temperature turbine components. For this cause, the discrete film holes are still the primary film-cooling configuration used on the hot gas path surfaces of the turbine up to now. Vast investigations have addressed that two mainly inherent drawbacks are associated with the discrete film holes. Firstly, the mutual interaction between coolant jet and primary flow induces a pair of counter rotating vortices with large scale, referred as kidney vortices. Secondly, the coolant jet is usually not capable of covering the zone between adjacent film holes. In order to improve the film cooling potential, considerable efforts have been devoted to overcome the above drawbacks in past decades. One major approach has been realized by the incorporation of exit shaping to the film holes [1, 2].

Over the past decades, the effects of shaped-hole geometries on the film cooling performance were extensively investigated, especially for the fan-shaped film cooling holes. Due to the anti-kidney vortices effect of diffused exit, the fan-shaped film-cooing hole leads to less shear mixing of the coolant jet with the primary flow, less penetration of the coolant jet into the primary flow, and wider lateral-coverage of the coolant jet related to the conventional or cylindrical film-cooling hole [3–5]. Recently, some innovative shaped holes were continually proposed and investigated, such as arrowhead-shaped hole (Okita and Nishiura [6]), branch or sister-shaped hole (Ely and Jubran [7], Abdelmohiman [8]), dumbbell-shaped and bean-shaped hole (Liu et al. [9], Kim et al. [10]), crescent-shaped hole (Kim et al. [10], Dai and Lin [11]), trench-shaped hole (Baheri et al. [12], Kalghatgi and Acharya [13]), NEKOMIMI and ridge-shaped hole (Kusterer et al. [14], Yang and Zhang [15]), tripod-shaped hole (Ramesh et al. [16]) and converging-slot hole (CONSOLE) (Sargison et al. [17, 18]), etc. Among these shaped holes, the converging-slot hole is of specific features, which is designed to offer the advantages of slot jet flow, while maintaining a structural strength similar to discrete holes. In this geometry, the coolant flow passage inside film cooling hole is turned gradually from an initially round cross section at the inlet to a finally rectangular slot at the exit with convergence in the axial direction and divergence laterally. Early investigations by Sergison et al. [17, 18] demonstrated clearly the advantages of converging-slot hole on film cooling performance with high film cooling effectiveness, uniform lateral cooling and low aerodynamic loss. Later, a lot of studies were devoted to the detailed flow field and film cooling performance of converging-slot hole [19–24]. It was interesting to find that the interaction between the coolant jet from inclined converging-slot hole and the primary flow results in a pair of counter rotating vortices originates from the edge of slot with a sense of rotation opposite to the kidney pair of conventional hole, being of great potential in the film cooling enhancement, especially under high blowing ratios. The above investigations also showed that properly changing the geometry parameters of converging slot-hole could improve the film cooling performance.

In a common sense, as the mutual interaction between coolant jets and primary flow is the dominant mechanism on the film cooling performance, the film cooling effectiveness of shaped holes is strongly affected by a large number of geometric parameters. Optimizing the shaped-hole geometry is an important issue for achieving the most favourable film cooling performance. Lee and Kim [25–27] performed a series of investigations to evaluate the effects of geometric variables of a laidback fan-shaped hole on the film-cooling effectiveness using a Reynolds-averaged Navier-Stokes analysis, and to optimize the shape of the hole using various surrogate models with the objective to maximize spatially averaged film cooling effectiveness. Recently, Naghashnejad et al. [28] made numerical simulations for the effects of interest parameters on the film cooling performance through a single row of inclined cylindrical holes. A predictive model was implemented by using GMDH-type (group method of data handing) neural networks to successfully present a correlation of area-weighted average adiabatic film cooling effectiveness. Chen et al. [29] performed a systematic study to investigate the combined effects of film-hole geometry, blowing ratio, density ratio, and freestream turbulence intensity on a flat-plate film cooling. Detailed film cooling effectiveness was obtained using a pressure-sensitive-paint technique. Wang et al. [30] made an optimization of a fan-shaped hole to improve film cooling performance by using radial basis functions neural network (RBF-NN) and genetic algorithm (GA). Three geometric parameters, including incline angle, lateral expansion angle and length of film cooling hole, were selected as the design parameters. The optimal geometric parameters were found under two typical blowing ratios of $M = 0.5$ and $M = 1.5$ respectively.

As mentioned above, the converging-slot hole is well demonstrated to be an ideal film-hole shape, however, to our knowledge, little efforts has been devoted to the effects of geometric factor of converging-slot-hole on the film cooling performance. In real, when the slot area at the film hole-exit is larger than that of round cross section at the inlet, the film hole will take on a diffusing but not diffusing feature. In the current study, an effort is devoted on the shape-optimization of round-to-slot holes for achieving the optimal adiabatic film cooling effectiveness by using CFD analysis and surrogate approximation methods. In order to illustrated the difference of optimal round-to-slot film cooling hole shapes under different

blowing ratios, two typical blowing ratios (such as M = 0.5 and M = 1.5) are taken into consideration for a row of film cooling holes on the flat surface.

2 Reference case and CFD validation

2.1 Reference case

For the purposes of the current study, the experimental model of film cooling on a flat plate with a row of inclined converging-slot holes presented by Yao and Zhang [31] is selected as a reference case for the optimization and numerical simulation validation. Considering that the film cooling issuing from a row of film cooling holes is of periodic feature, only one hole-to-hole pitch containing a single hole is selected as the computational domain by treating the two lateral boundaries as the symmetric planes, as schematically shown in Fig. 1a. The computational domain is composed of three sub-zones, namely, the primary flow passage, film hole, and cooling air channel. Also, as the current study concerns on adiabatic film cooling effectiveness, the perforated wall is not taken into consideration in the computational domain. The film hole is located at 20*d* downstream of the primary flow passage inlet and 16*d* upstream of the primary flow passage outlet, as seen in Fig. 1b. The diameter of film cooling hole (*d*) is fixed as 5 mm and the hole-to-hole pitch (*P*) is fixed as 2.4*d*. The height of primary flow passage (in *y*-direction) is chosen as 12*d*. The secondary or coolant flow passage has a height of 4*d*. Its streamwise extent is set as $-15 \le x/d \le 5$. Two symmetric planes are applied to the domain boundary. The coordinate origin is located at the center of film-hole exit.

The round-to-slot hole is schematically shown in Fig. 1c. Its cross-section changes from an initially round cross section at the inlet to a finally rectangular slot at the exit within a hole-length (*L*). In side view the walls of the round-to-slot hole converges and in top view the walls diverge. Given a fixed film cooling hole diameter and hole-to-hole pitch (*P*), the other main geometric parameters of a round-to-slot hole are also outlined in Fig. 1c, including inclination angle (θ), exit slot length (*l*) and width (*s*), as well as the height of film hole (*t*). For the current reference case, the inclination angle of film cooling hole (θ) is 35° in the streamwise direction or *x*-direction. The film-hole height or perforated plate thickness (*t*) is 6 mm, giving a height-to-diameter ratio (*t*/*d*) of 1.2. The slot length (*l*) and width (*s*) at film-hole exit are 10 mm and 1 mm respectively, giving a slot aspect (*l*/*s*) of 10.

2.2 Computational scheme

Three-dimensional numerical simulation is employed by using Fluent-CFD software [32]. A realizable k-ε turbulence model provided in the software is used as the turbulence closure and the near wall region is modeled using enhanced wall treatment. According to the work of Harrison and Bogard [33], as the realizable model is designed with the intent of offering viable solutions for jet-in-crossflow interaction by modifying the dissipation rate and eddy-viscosity equations, it offers fairly good predictions of film cooling performance for a two-equation model. This turbulence model has been successfully used by some researcher (such as Yao et al. [23, 24], Silieti et al. [34], Ely and Jubran [7]). Convergence is achieved when both of the following criteria

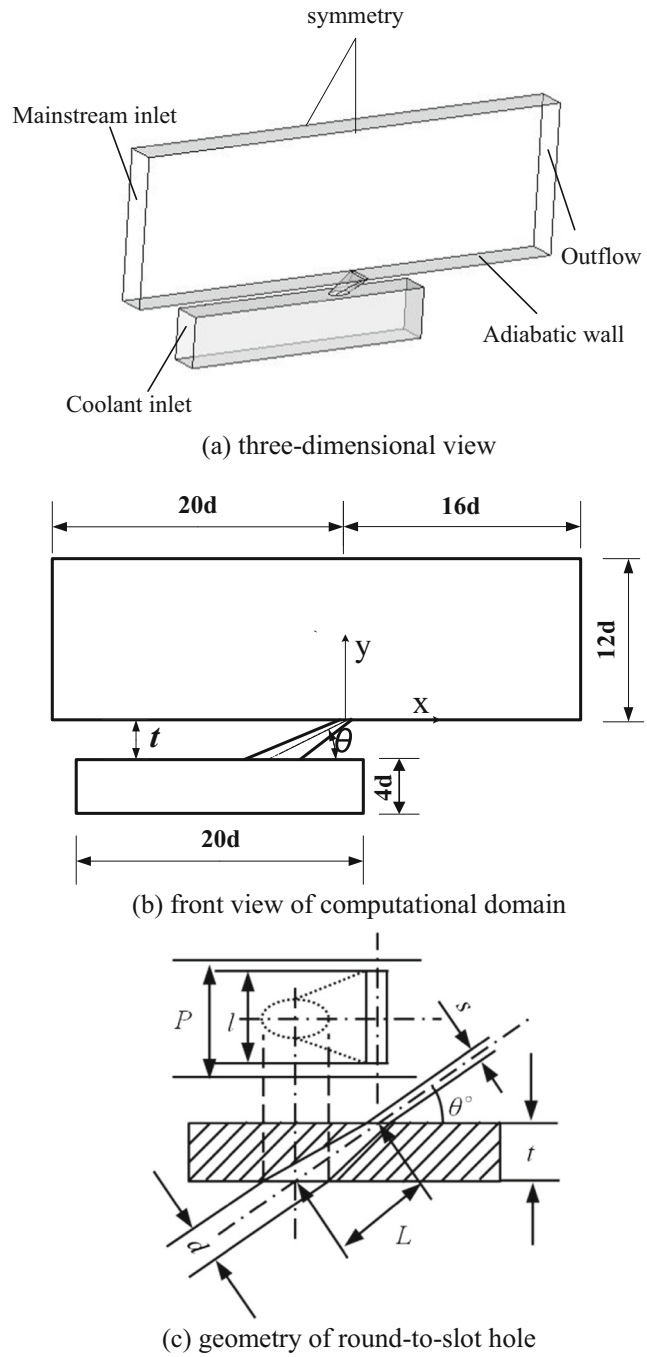


Fig. 1 Schematic of computational model

have been met: (a) reduction in all residuals of five orders of magnitude, and (b) the temperature change not exceeding ±0.1 K in local temperature prediction on the protected surface for an additional 30 iterations.

Overall multi-block grids are generated for the whole computational domain, as shown in Fig. 2. The hexahedral structured grids are applied for both the primary flow and the secondary flow channels. While for the round-to-slot hole, the tetrahedral unstructured grids are adopted. To achieve a grid independent solution, four sample grid systems are designed. Their total numbers of grids are 513,332, 908,074, 1,304,068 and 2,001,274 respectively. Figure 3 presents the influence of grid number on the adiabatic film cooling effectiveness distribution at film-hole centerline for a typical condition. It is shown that the computational results are not sensitive to the grid number when the grid number is larger than 1,304,068. As a consequence, a grid system with approximate 1.3 million grids is finally chosen as the grid independent solution for subsequent simulations since the grid-independent result is successfully obtained.

In the computations, both the primary flow and the secondary flow are treated as ideal air. In Fluent-CFD software, the physical parameters of are set as follows: the ideal-gas based density; Kinetic-theory based specific heat and thermal conductivity; Sutherland law based viscosity.

The boundary conditions of computational domain are specified as the followings.

The primary flow inlet is defined as a velocity inlet with a mean inlet velocity of 25 m/s. The temperature of mainstream is fixed as 335 K. A turbulence intensity of 2% and a turbulence length scale of 3% inlet height in the primary flow passage are used. The flow outlet is set as pressure outlet with static pressure of 101,325 Pa.

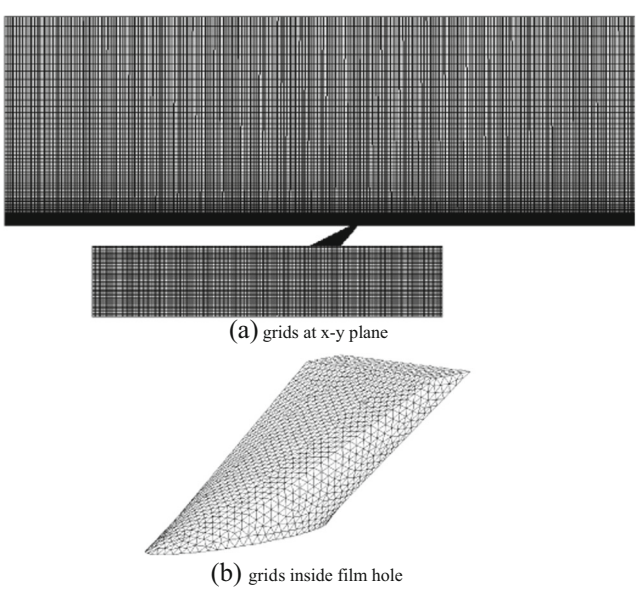


Fig. 2 Computational grids

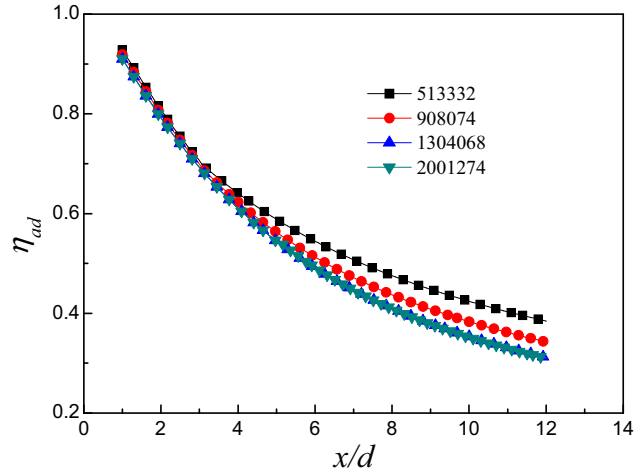


Fig. 3 Effect of grid number on adiabatic film cooling effectiveness at film-hole centerline

The coolant air is introduced into film holes from a coolant channel by using the mass flow inlet. The mass flow is determined according to the blowing ratio. A turbulence intensity of 2% and a turbulence length scale of 3% inlet height of the secondary flow passage are used. The coolant temperature is set as 300 K.

For the round-to-slot hole, it is noted that the cross-sectional area or the coolant flow velocity varies from the hole inlet to the hole exit. According to the previous works of Yao and Zhang [31], the coolant parameters used in the definition of blowing ratio are taken from the inlet of film hole.

$$M = \frac{(\rho_c u_c)_{hole-inlet}}{\rho_\infty u_\infty} \tag{1}$$

where ρ_c and u_c are density and velocity of the secondary flow or coolant flow at inlet of film hole, respectively. ρ_∞ and u_∞ are density and velocity of the primary flow respectively.

The symmetric boundary conditions are applied for the two symmetric boundaries of the computational domain. The symmetry planes are modeled adiabatically. Each plane is given zero velocity in the spanwise direction (z-direction), thus preventing the flow from crossing the boundary but yet allowing a velocity profile to develop. In terms of the boundary conditions on the solid walls, thermally adiabatic and no-slip conditions are used.

The local and laterally-averaged adiabatic film cooling effectiveness is calculated according to Eqs. (2) and (3), respectively, wherein the numerical integration of local adiabatic film cooling effectiveness within one hole-to-hole pitch ($P = 2.4d$) is used to find the laterally-averaged effectiveness values.

$$\eta_{ad} = \frac{T_\infty - T_{aw}}{T_\infty - T_c} \tag{2}$$

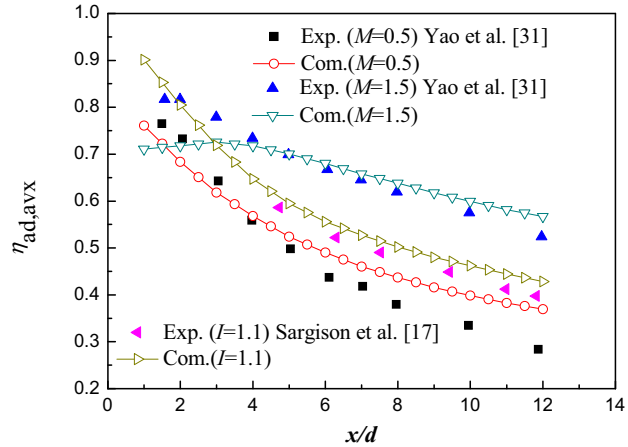
$$\eta_{ad,avx} = \frac{1}{2.4d}\int_{-1.2d}^{1.2d}\eta_{ad}dy \quad (3)$$

where T_c is coolant flow temperature, T_∞ is primary flow temperature, T_{aw} is adiabatic wall temperature at the plate surface suffering the primary flow.

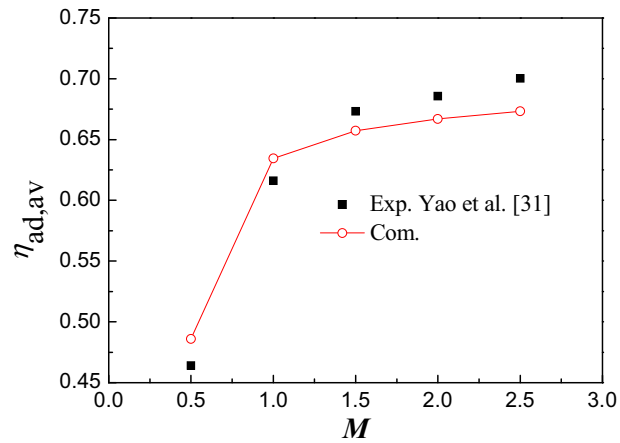
2.3 CFD validation

For the validation purpose, two examples are presented here, the corresponding computational models are built up in accordance with the corresponding test model.

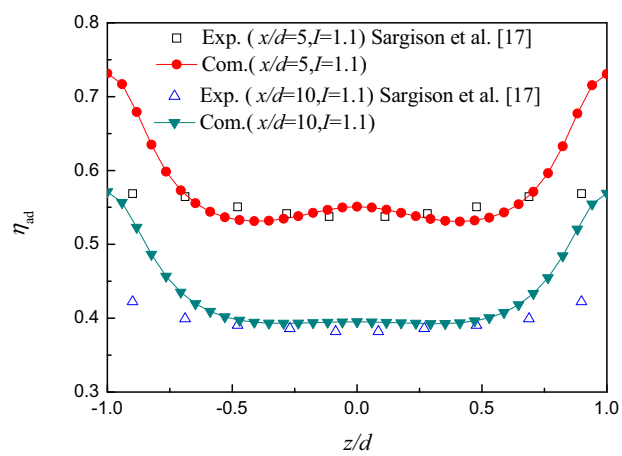
Figure 4a, b present the comparison of current simulation and experimental data of Yao et al. [31]. In the viewing of laterally-averaged adiabatic film cooling effectiveness distribution, it is seen from Fig. 4a that the computed laterally-averaged adiabatic film cooling effectiveness near the film-hole outlet is a little under-estimated. While for the region apart from film-hole outlet, the computed laterally-averaged adiabatic film cooling effectiveness is a little over-estimated. The maximum reletive deviation is appeared near the film cooling hole under a high blowing ratio, which is about 15%. The main causes for this deviation between the present simulation and the previous test by Yao and Zhang [31] are suggested to be the following three aspects. Firstly, it is noted that the tested data are measured in a thin plate in which the thermal conductive effect is generally existed although the solid wall has a little thermal conductivity. While in the current numerical simulation, a perfect adiabatic thermal boundary is used. Secondly, the 'real' converging-slot hole shape (especially the transition contour) of test model is not completely the same as that in the computation due to the manufacture imperfection. Thirdly, as previously by many researchers, the coolant flow at the entrance and inside the fiom cooling hole is very complicated. Especially, under a high blowing ratio, the coolant jet with a strong momentum flux ratio generally produces separation bubble near the film-cooling-hole outlet. In this situation, the numerical simulation precision is affected significantly by the turbulence model. Figure 4b shows the variation of the spatially-averaged adiabatic film cooling effectiveness (ranging between $x/d = 2$ and $x/d = 12$) with the blowing ratio. For the converging-slot hole, as the blowing ratio is increased, the spatially-averaged adiabatic film cooling effectiveness is increased rapidly under low blowing ratios. As the converging-slot hole has the ability for suppressing coolant jet penetration into the primary flow, the increase of blowing ratio makes the streamwise momentum of coolant jet be enhanced, thus leading to a strong film cooling capacity downstream the film hole. In the viewing of spatially-averaged adiabatic film cooling effectiveness, the maximum reletive deviation is certainly less than 8%. Figure 4a also presents the comparison of laterally-averaged adiabatic film cooling effectiveness distribution between the



(a) laterally-averaged adiabatic film cooling effectiveness



(b) spatially-averaged adiabatic film cooling effectiveness



(c) Local adiabatic film cooling effectiveness in the lateral direction

Fig. 4 Comparison of current simulation and experimental data of Yao et al. [31] and Sargison et al. [17]

current simulation and the experimental results presented by Sargison et al. [17]. Figure 4c shows the comparison of local adiabatic film cooling effectiveness in the lateral direction under the momentum flux ratio (defined as $\rho_c u_c^2/\rho_\infty u_\infty^2$) of $I = 1.1$. It is seen that the deviation between the computed and

tested local cooling effectiveness is very small at the center region but a little higher at the side region. In general, the current numerical simulation is reasonable.

3 Optimization methodology

3.1 Design variables and objective function

As illustrated in Fig. 1c, six geometric parameters are involved in round-to-slot holes, such as film cooling hole diameter (d), inclination angle (θ), exit slot length (l) and width (s), height of film hole (t), as well as hole-to-hole pitch (P). In the current study, film cooling hole diameter (d), exit slot length (l) and hole-to-hole pitch (P) are fixed. The other geometric parameters are selected as the design variables. All the geometric parameters are summarized in Table 1. For such arrangements, when s/d is equal to 0.392, the exit area of slot is the same as that of an initially round cross section at the inlet. Therefore, the round-to-slot hole with s/d<0.392 is a converging slot-hole where the cross-sectional area decreases from the inlet to the outlet and the cooling air accelerates inside the film cooling hole. While the round-to-slot hole with s/d>0.392 is a diffusing slot-hole where the cross-sectional area increases from the inlet to the outlet and the cooling air decelerates inside the film cooing hole.

A single-objective optimization problem is taken into account in the current study. The spatially-averaged film-cooling effectiveness is selected as the objective function to be maximized. According to previous works [25–27], the spatially averaged film-cooling effectiveness is generally determined in a specific zone with one hole-to-hole pitch in the lateral width and 10d~20d in the streamwise length. For the current study, the spatially averaged film-cooling effectiveness defined in Eq. (4) is adopted wherein the numerical integration of laterally-averaged adiabatic film cooling effectiveness between $x/d = 2$ and $x/d = 12$ is used to find the spatially-averaged effectiveness value.

$$\eta_{ad,av} = \frac{1}{12d}\int_{2d}^{12d} \eta_{ad,avx} dx \tag{4}$$

Table 1 Fixed geometric parameters and design variables

Fixed geometric parameters	Symbol	Value	
Hole diameter	d	5 mm	
Hole-to-hole pitch	P	12 mm (2.4d)	
Exit slot length	l	10 mm (2d)	
Design variable	Symbol	Lower bound	Upper bound
Exit slot width	s	0.5 mm (0.1d)	3.5 mm (0.7d)
Film hole height	t	2.5 mm (0.5d)	20 mm (4d)
Inclination angle	θ	20°	80°

The definition function of the current optimization problem is expressed as:

$$\min F(s,t,\theta) = \frac{1}{\eta_{ad,av}}$$
$$s.t.\begin{cases} s_{\min} \le s \le s_{\max} \\ t_{\min} \le t \le t_{\max} \\ \theta_{\min} \le \theta \le \theta_{\max} \end{cases} \tag{5}$$

where $F(s,t,\theta)$ is the fitness function.

3.2 Optimization scheme

Figure 5 describes the surrogate based optimization procedure used in the present work.

A surrogate model is firstly constructed and validated. The surrogate modeling technique is one of optimization methods, which has been widely used in engineering applications due to its intrinsic advantages. In the present study, radial basis function neural network (RBFNN) is applied for constructing the surrogate model, which is originated from the well-studied subject of multivariate regularization theory, providing fast convergence speed for hyper-surface reconstruction coupled with efficient noise elimination [35]. In the present study, RBFNN is established by a customized command 'newrbe' in Matlab software. For the purpose of RBFNN construction, 25 design points are selected as training samples with the help of Latin Hypercube Sampling (LHS) and 8 design points are selected as testing samples randomly within the design scope of design variables, as illustrated in Tables 2 and 3 respectively. Correspondingly, a series of computations are performed to determine the objective function values under two typical

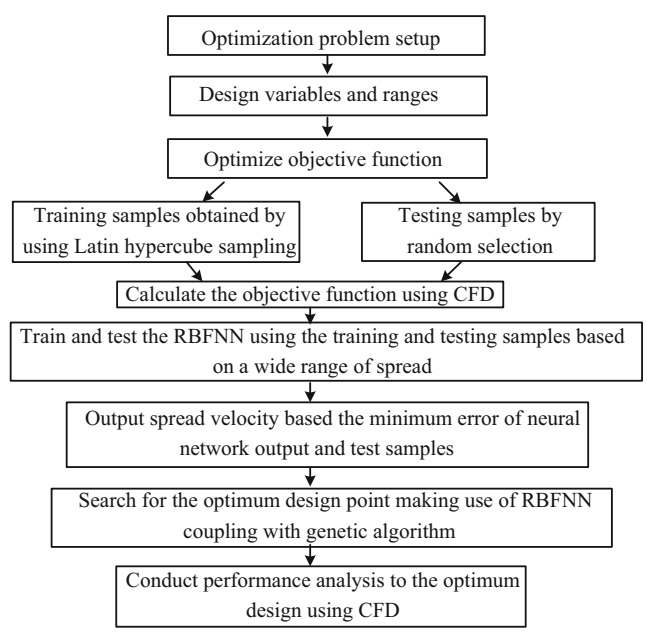


Fig. 5 Surrogate based optimization procedure

Table 2 Training and testing samples

No.	s(mm)	t(mm)	θ(°)	$\eta_{ad,av}$ ($M=0.5$)	$\eta_{ad,av}$ ($M=1.5$)
1	0.5	3	20	0.491	0.659
2	0.5	9	40	0.447	0.523
3	0.5	15	60	0.410	0.397
4	0.5	20	80	0.335	0.360
5	1	5	30	0.487	0.680
6	1	12	50	0.456	0.566
7	1	19	70	0.418	0.506
8	1.5	3	40	0.480	0.737
9	1.5	9	20	0.478	0.762
10	1.5	15	80	0.427	0.586
11	1.5	20	60	0.439	0.590
12	2	5	50	0.479	0.735
13	2	12	70	0.437	0.656
14	2	19	30	0.461	0.703
15	2.5	3	60	0.456	0.694
16	2.5	9	80	0.421	0.504
17	2.5	15	20	0.464	0.787
18	2.5	20	40	0.461	0.679
19	3	5	70	0.425	0.622
20	3	12	30	0.454	0.792
21	3	19	50	0.445	0.728
22	3.5	3	80	0.352	0.287
23	3.5	9	60	0.430	0.755
24	3.5	15	40	0.449	0.757
25	3.5	20	20	0.461	0.761

blowing ratios of $M = 0.5$ and $M = 1.5$. The calculated results are also illustrated in the above tables. Figure 6 shows the comparison between the RBFNN output and the CFD calculation results for the testing samples. At $M = 0.5$, the relative error is less than 2%. At $M = 1.5$, the relative error is less than 4%. In general, the RBFNN outputs are in good agreements with the CFD calculation results.

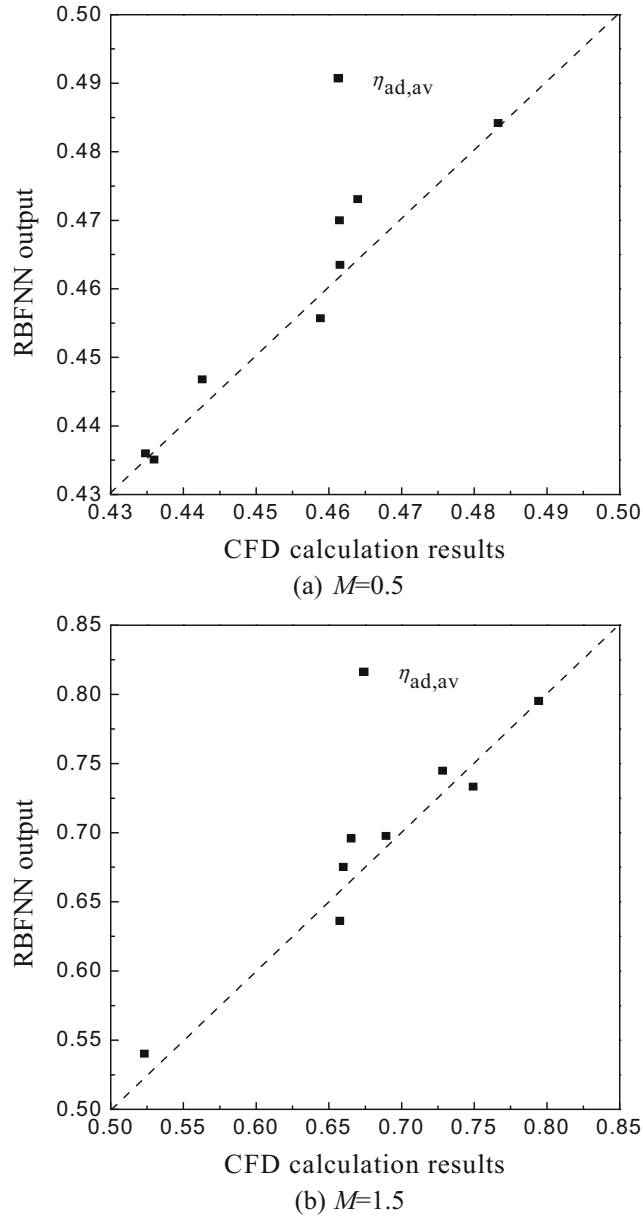


Fig. 6 Comparison of RBFNN output and CFD calculation results

Table 3 Testing samples

No.	s(mm)	t(mm)	θ(°)	$\eta_{ad,av}$ ($M=0.5$)	$\eta_{ad,av}$ ($M=1.5$)
1	0.9	4	30	0.483	0.665
2	1.3	7	60	0.459	0.658
3	1.6	9	50	0.464	0.689
4	2.1	10	40	0.461	0.728
5	2.4	13	70	0.443	0.660
6	2.7	17	80	0.435	0.523
7	3.1	6	30	0.462	0.754
8	3.4	8	60	0.436	0.749

Once the surrogate model is constructed by RBFNN, the optimum global point is searched by the genetic algorithm which involves in a process for function optimization that mimics the genetic reproduction process experienced by natural evolution organism [36, 37]. The general flow chart of searching for the optimal global point by genetic algorithm is well described in the work of Wang et al. [30].

Finally, the film cooling process corresponding to the optimum global point is numerically analyzed. The improvement of adiabatic film cooling effectiveness related to the reference case is illustrated and the inherent mechanism is outlined.

4 Results and discussions

In this section, the optimization results are presented and the influence of optimal round-to-slot geometry on film cooling performance is illustrated.

4.1 Optimization results

The optimization results under two typical blowing ratios of $M = 0.5$ and $M = 1.5$ are shown in Tables 4 and 5, respectively. In these tables, the spatially-averaged adiabatic film cooling effectiveness for the reference case is also listed. For the optimization case, both the RBFNN outputs and CFD values are presented for comparison.

As seen in Table 4, the optimal geometric parameters of (s,t,θ) is determined as (1.1 mm, 3 mm, 20°) under a low blowing ratio of $M = 0.5$. By comparing with the reference case, it is found that the optimal round-to-slot hole has a little wide slot width. But the film-hole height and inclination angle are significantly less than the corresponding values of the reference case. It is also found that the spatially-averaged film cooling effectiveness after optimization is only about 4% increase related to the reference case. Under a low blowing ratio, the optimal round-to-slot hole remains the converging feature.

While under a high blowing ratio of $M = 1.5$, as seen in Table 5, the optimal spatially-averaged film cooling effectiveness is obviously improved in comparison with the reference case, with approximately 23% increase. In this situation, it is found that the optimal round-to-slot hole is of diffusing feature with a wider slot-width of 2.2 mm related to the reference case. The other design variables, such as the film-hole height and inclination angle, are little less than the corresponding values of the reference case.

4.2 Detailed film cooling behaviors

Figure 7 presents the dimensionless temperature contours on some cross-sectional planes along streamwise direction. Figure 8 presents the dimensionless temperature distribution on the film-hole centerline plane at $z/d = 0$. Here the dimensionless temperature is defined as $\Theta = \frac{T-T_c}{T_\infty - T_c}$.

Under $M = 0.5$, it is found that the optimal round-to-slot hole seems has nearly the same temperature contours as the reference case on the cross-sectional planes along streamwise direction, as seen in Fig. 7a, b. By comparing dimensionless temperature distribution on the film-hole centerline plane, as seen in Fig. 8a, b, it is found that the optimal round-to-slot hole produces a thicker and longer film-layer coverage than the reference case near the film-hole exit, especially between $x/d = 0$ and $x/d = 2$. Referring to Table 4, the optimal round-to-slot hole has a little wide slot width in comparison with the reference case, both having converging feature. For the converging round-to-slot hole, the coolant jet issuing from the film hole exit ($u_{c,jet}$) is significantly greater than that at the film hole inlet due to the acceleration of coolant flow inside the film hole, however, the normal penetration of coolant jet into the primary flow could be effectively suppressed by its specific mechanism, as seen in Fig. 9. A pair of counter rotating vortices originates at the side-edges of slot, which has a sense of rotation opposite to the kidney vortex pair as seen in the conventional cylindrical hole. This pair of counter rotating vortices makes the primary flow to be suctioned into the film layer form the film-hole center. In addition, two small corner vortices are also formed at both sides of the slot. Obviously, the formation of these special vortices is dominated by the specifically shaped hole which makes the coolant to flow laterally toward to the narrow sides of slot inside the round-to-slot hole. Therefore, by comparing the optimal and reference holes, the former has a shorter transition length from initial round to final slot ($t/\sin(\theta) = 8.77$ mm) than the latter case ($t/\sin(\theta) = 10.7$ mm), which makes the lateral flow of coolant inside the round-to-slot hole more stronger and consequently improves the lateral spreading of coolant jet downstream the film-hole exit. This flow feature is directly observed in Fig. 10, which is drawn on the x-z plane tightly closed to the wall surface. Besides, the reduction of film-hole inclination angle is helpful for transferring more coolant jet momentum to the streamwise direction.

With regard to the high blowing ratio of $M = 1.5$, the shape optimization of round-to-slot hole is significantly behaved on the improvement of film cooling. As seen in Fig. 7c, d, the optimal round-to-slot hole changes the temperature contours obviously on the cross-sectional planes along streamwise direction in relative to the reference case, producing a more-like lobe-shaped temperature contour with the lobe valley corresponding to the film-hole center. Also, the temperature

Table 4 Comparison of reference case and optimization case under $M = 0.5$

	Design variables			$\eta_{ad,av}$		Relative error
	s (mm)	t (mm)	θ (°)	CFD	RBFNN	
Reference	1	6	35	0.48		
Optimization	1.1	3	20	0.5	0.51	2%

Table 5 Comparison of reference case and optimization case under $M = 1.5$

	Design variables			$\eta_{ad,av}$		Relative error
	s (mm)	t (mm)	θ (°)	CFD	RBFNN	
Reference	1	6	35	0.66		
Optimization	2.2	3.9	29.1	0.81	0.82	1.2%

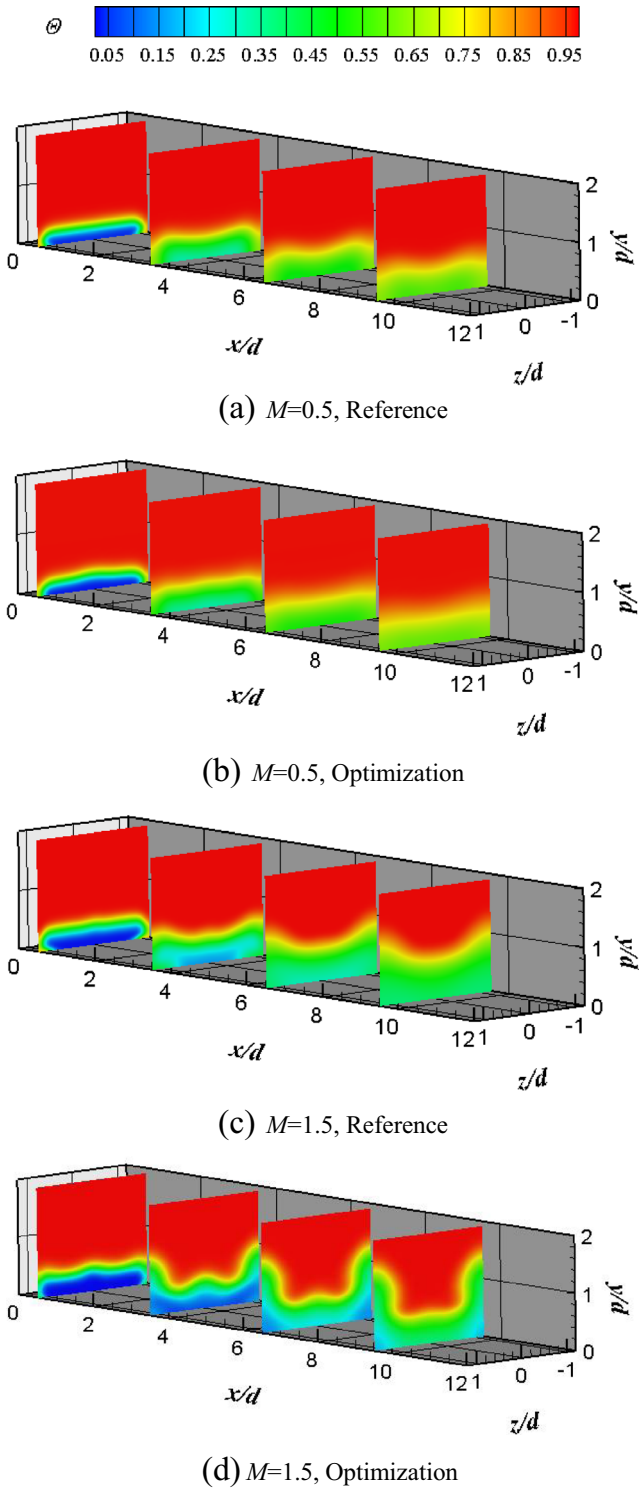


Fig. 7 Dimensionless temperature contours on some cross-sectional planes along x direction

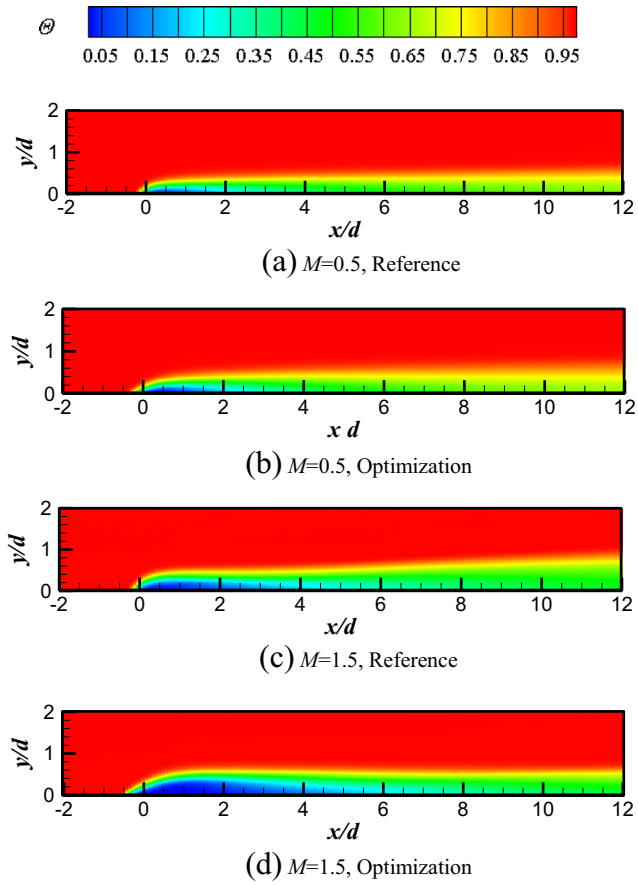


Fig. 8 Dimensionless temperature contours on film-hole centerline plane

distribution is improved significantly for the optimal film hole between $x/d = 0$ and $x/d = 6.5$.

The most inspiring finding is that the optimal round-to-slot hole is of diffusing feature under a high blowing ratio, which is completely contrary to the low blowing situation. As the increase of blowing ratio, the coolant flow mean velocity inside the film-hole is increased. For the reference case or a converging round-to-slot film cooling hole, the transition section varies rapidly with a great lateral-divergence angle to slot narrow side, witch maybe affect the internal flow inside the round-to-slot hole and consequently the mutual interaction between the coolant jet and the primary flow. As seen in Fig. 11a, the lateral coolant jet is found to be left off the protected surface and the corner vortices at both sides of the slot seems to have a bigger scale. While for the optimal round-to-slot hole, as the slot area at the film-hole exit is larger than the inlet area, the transition section varies smoothly form the round inlet to slot exit, which is supposed to produce a more reasonable interaction between the coolant jet and the primary flow, as seen in Fig. 11b. The corner vortices between two adjacent film-holes are effectively reduced and the film layer coverage in the lateral direction is improved in comparison with the reference case.

Figure 12 shows the velocity vector and lateral-velocity contours on the x-z plane tightly closed to the wall surface under $M = 1.5$. For the reference case or a converging round-to-slot film cooling hole, as seen in Fig. 12a, the lateral flow of the coolant jet is mainly displayed in the vicinity of the slot narrow end. Behind the slot exit, a pair of normal vortices is

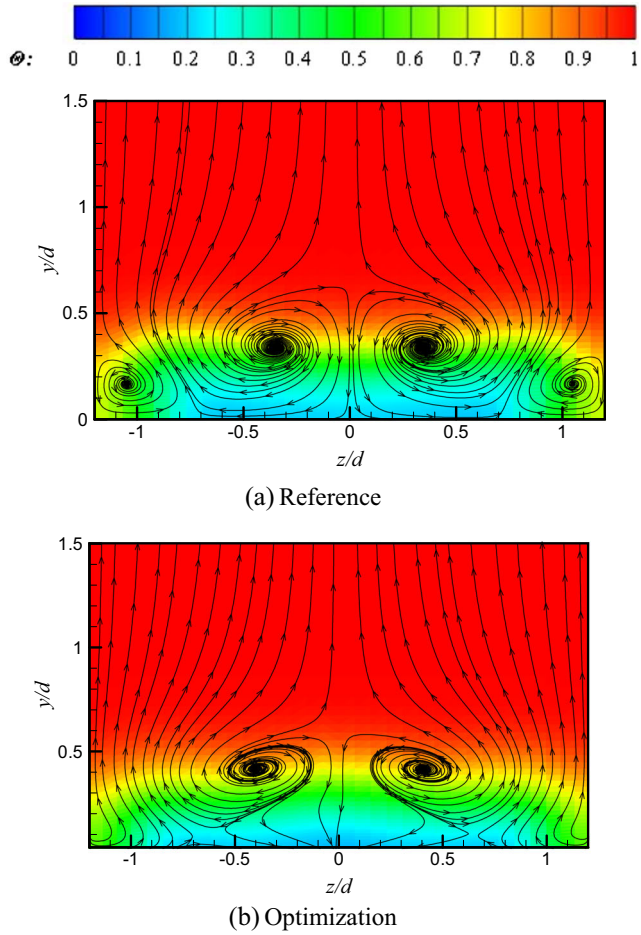


Fig. 9 Streamline and dimensionless temperature distribution on the normal plane at $x/d = 2$ under $M = 0.5$

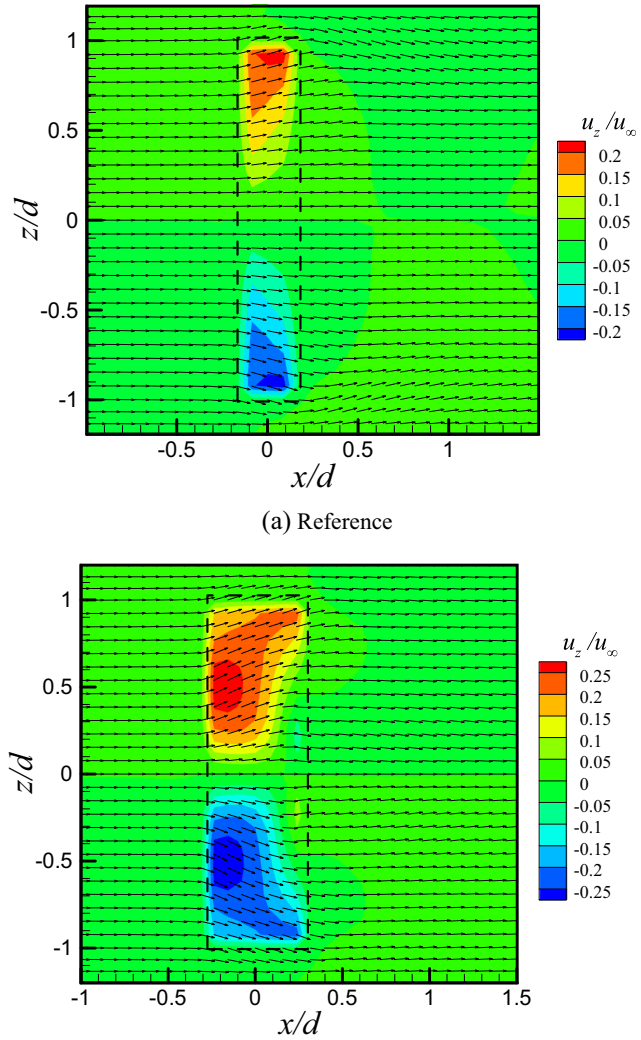


Fig. 10 Velocity vector and lateral-velocity contours on x-z plane closed to the wall under $M = 0.5$

displayed near the film-hole centerline. Combined with the finding of lateral jet lift off the wall as displayed in Fig. 11a, it is inferred that the film layer coverage in the lateral direction is not the most reasonable for the converging round-to-slot film cooling hole. By shape optimization of the round-to-slot hole, as seen in Fig. 12b, the lateral spreading of coolant jet is significantly improved in comparison with the reference case.

In order to display the effects of shaped-holes optimization on the film cooling performance more intuitively, the vortex structures under the blowing ratio of $M = 1.5$ are presented in Fig. 13. These vortex structures are represented by iso-surfaces of swirling strength colored by ω_x. From this figure, the horseshoe vortices are displayed clearly around the leading edge of the film holes. When the primary flow progresses through the film-cooling hole, the coolant jet plays an acttion of similar blockage to the cross flow. Due to the larger ejecting velocity of the coolant jet for the reference case, the horseshoe vortice around the leading edge of the film cooling hole has a wider size than the optimation case. On the slot sides, the corner vortices are formed immidiately downstream of the film hole exit. It is apparent the corner vortices for the reference case have a relatively longer extent than the optimization case. At the top side, the vaule of streamwise vortice is negative, indicating a sense of counter-clockwise rotation. While at the bottom side, the case is opposite. Of particular significance to the vortex structures near the film-hole exit is that the optimization film-hole produces obviously longer extent in comparison with the reference case. Beside, the the vaule of streamwise vortice is positive at the top side and negative at the bottom side, suggesting that the lateral flow of coolant jet exiting from the optimization produces a stronger influence far from the film cooling hole exit.

Figure 14 shows the distribution of local adiabatic film cooling effectiveness on the protected surface before and after optimization. Figure 15 presents the local adiabatic film cooling effectiveness distribution in the lateral direction at some streamwise locations. Under $M = 0.5$, it is seen from Fig. 14a, b that the optimal round-to-slot hole produces a little improvement on the local adiabatic film cooling effectiveness

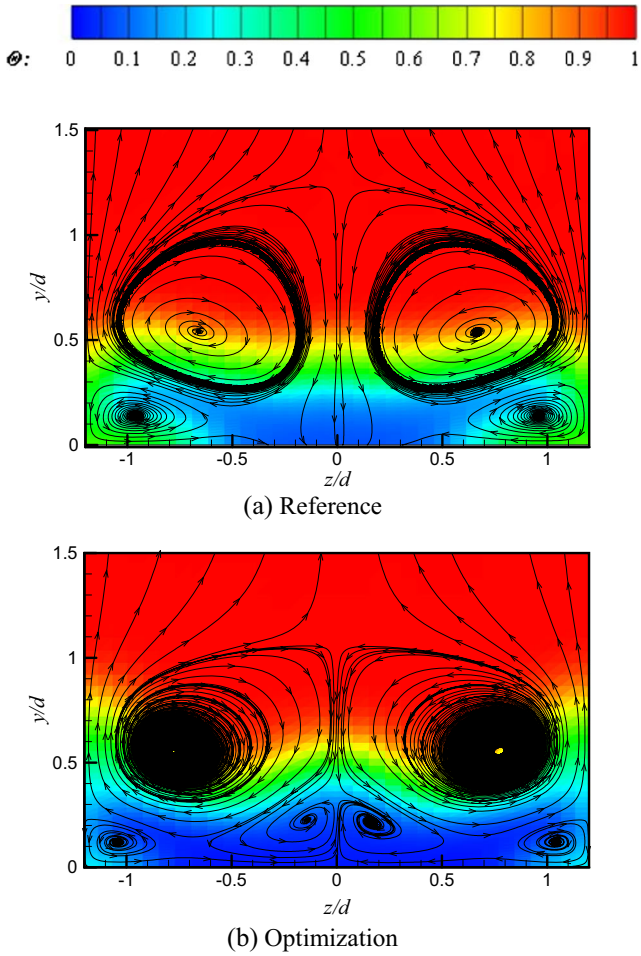


Fig. 11 Streamline and dimensionless temperature distribution on the normal plane at $x/d = 2$ under $M = 1.5$

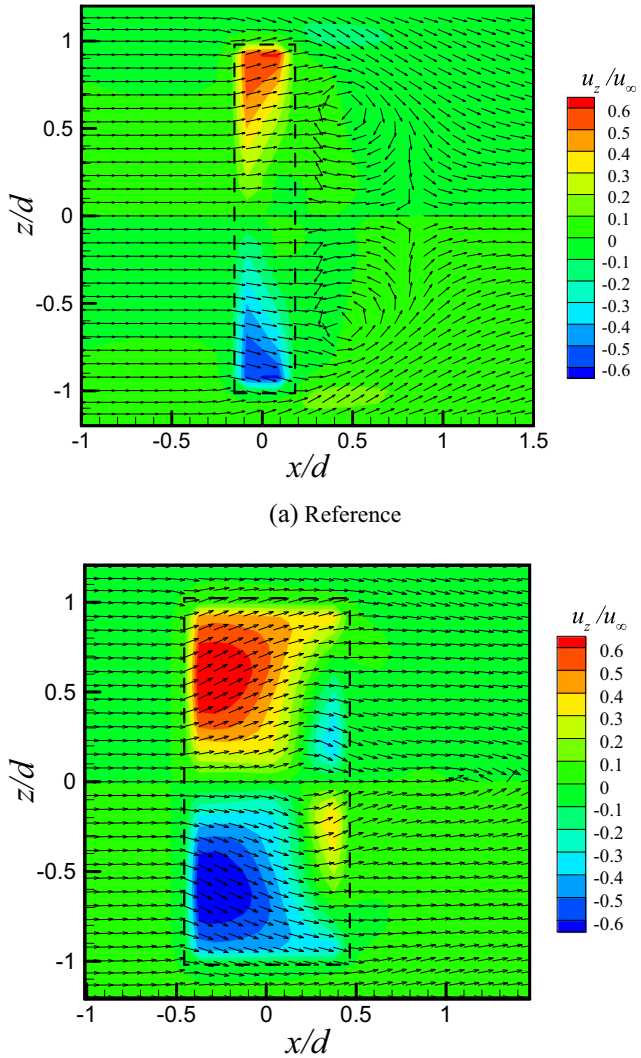
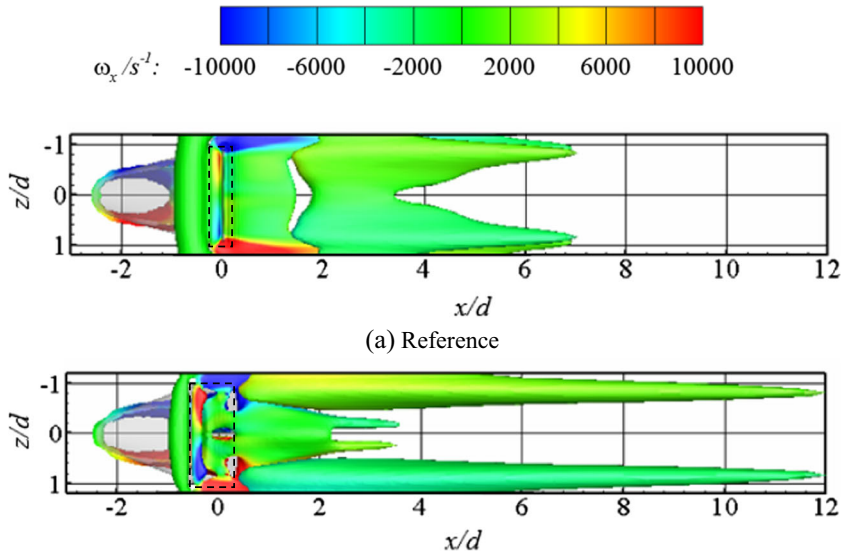


Fig. 12 Velocity vector and lateral-velocity contours on x-z plane closed to the wall under $M = 1.5$

distribution in comparison with the reference case. As shown in Fig. 15a, the local adiabatic film cooling effectiveness of optimal round-to-slot hole in the spanwise direction is less than the corresponding ones in some ranges of z/d but larger in the other ranges. Under $M = 1.5$, the local adiabatic film cooling effectiveness distribution on the protected surface is significantly improved by the round-to-slot hole optimization, as seen in Fig. 14c, d. At the streamwise location of $x/d = 4$, the optimal round-to-slot hole achieves a nearly uniform local adiabatic film cooling effectiveness distribution with a high η_{ad} value. However, for the reference case or a converging round-to-slot film cooling hole, the local adiabatic film cooling effectiveness is decreased rapidly from 0.8 at $z/d = 0.5$ to 0.6 at $z/d = 1.0$, as seen in Fig. 15b. Obviously, this rapid reduction of local adiabatic film cooling effectiveness is associated with the lateral jet lift off the wall. At the streamwise location far from the film-cooling-hole exit (such as $x/d = 12$), although the local adiabatic film cooling effectiveness achieved by the optimal round-to-slot hole in the region of $|z/d| \leq 0.5$ is almost the same as that of the reference case, the improvement of local adiabatic film cooling effectiveness in the region of $|z/d| \geq 0.5$ is still remarkable.

Figure 16 presents the laterally-averaged adiabatic film cooling effectiveness distribution. Under $M = 0.5$, as shown in Fig. 16a, the laterally-averaged adiabatic film cooling effectiveness tightly downstream the film-hole outlet is effectively enhanced. For instance, at $x/d = 2$, the laterally-averaged adiabatic film cooling effectiveness could be increased approximately 11% by the optimal film-hole in comparison with the reference case. When x/d is beyond 6, the difference of laterally-averaged film cooling effectiveness between the reference film hole and optimal film-hole tends to disappear. Under $M = 1.5$, as shown in Fig. 16b, a general 15%~30% increase of the laterally-averaged adiabatic film cooling effectiveness could be achieved by the film-hole shape optimization among $x/d = 2$ and $x/d = 12$. In general, the improvement



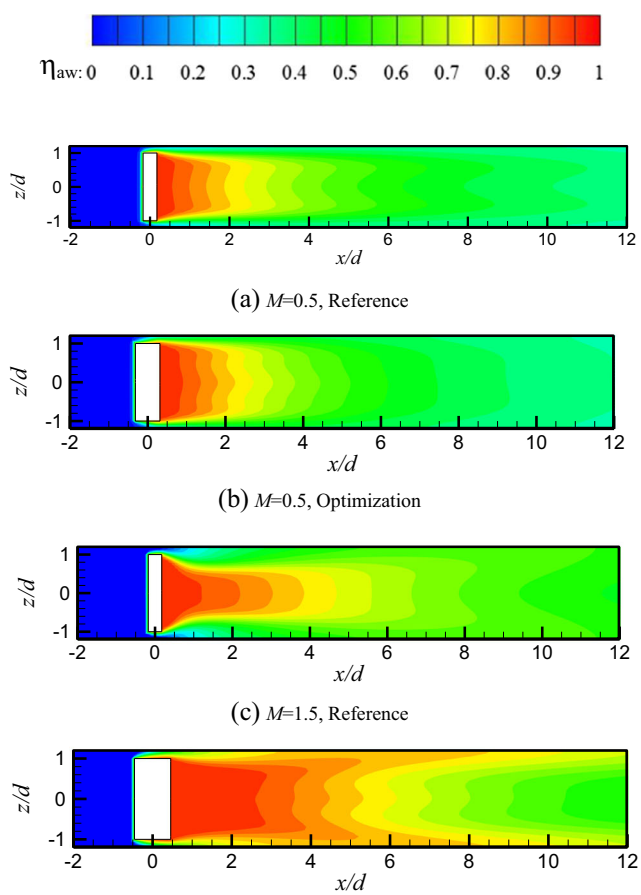
(a) Reference

(b) Optimization

Fig. 13 Vortex structures colored by ω_x under $M = 1.5$

of the laterally-averaged adiabatic film cooling effectiveness is degraded as the increase of x/d.

According to the previous works on the film cooling mechanism, low normal penetration and strong streamwise



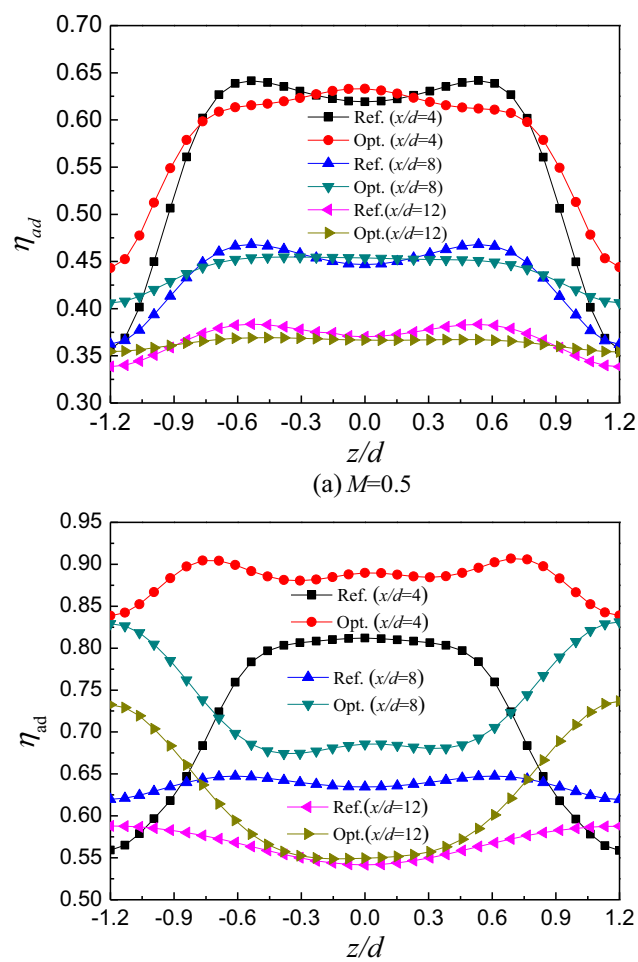
(a) M=0.5, Reference

(b) M=0.5, Optimization

(c) M=1.5, Reference

(d) M=1.5, Optimization

Fig. 14 Distribution of local adiabatic film cooling effectiveness



(a) M=0.5

(b) M=1.5

Fig. 15 Local adiabatic film cooling effectiveness in the lateral direction

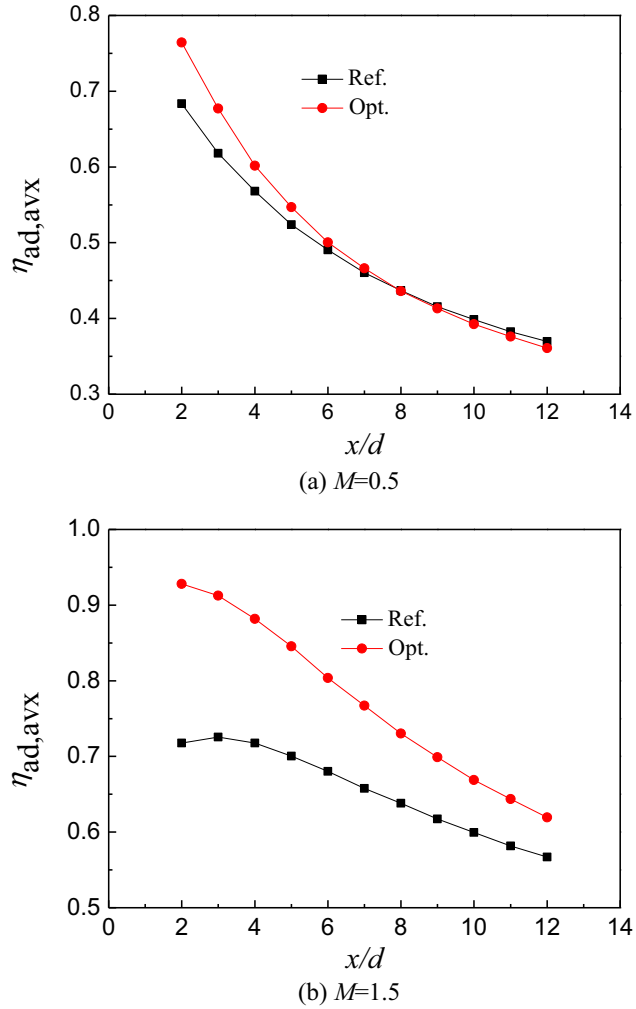


Fig. 16 Laterally-averaged adiabatic film cooling effectiveness

extension of the coolant jet flow are benefit to the enhancement of the film cooling effectiveness. Under a small blowing ratio, the ejection momentum of coolant jet is relatively weaker. In this situation, the converging round-to-slot hole is helpful for achieving stronger streamwise extension capacity than the diffusing round-to-slot hole. While under a large blowing ratio, the diffusing round-to-slot hole is more pronounced for suppressing the normal penetration of coolant jet into the primary flow. This competition of two roles determine the optimal round-to-slot hole shape under different blowing ratios, either of converging or diffusing feature.

5 Conclusions

This paper summarizes a single-objective optimization research for improving the adiabatic film cooling effectiveness of single row round-to-slot film cooling holes on a flat surface by using three-dimensional CFD analysis and surrogate approximation methods. Among the main geometric parameters, the dimensionless hole-to-hole pitch (P/d) and slot length-to-diameter (l/d) are fixed as 2.4 and 2 respectively and the other three parameters (such as slot width-to-diameter (s/d), hole height-to-diameter ratio (t/d) and inclination angle (θ)) are chosen as the design variables. The geometric shape optimization of round-to-slot film cooling holes is carried out under two typical blowing ratios of $M = 0.5$ and $M = 1.5$ by selecting a spatially-averaged adiabatic film cooling effectiveness between $x/d = 2$ and $x/d = 12$ as the objective function which is to be maximized. Radial basis function neural network (RBFNN) is applied for constructing the surrogate model and the optimal design point is searched by a kind of genetic algorithms. In the current study, the RBFNN output results are in good agreements with the spatially-averaged adiabatic film cooling effectiveness obtained from three-dimensional CFD analysis.

By optimization, different optimal geometric parameters are determined corresponding to the blowing ratio. Under a low blowing ratio of $M = 0.5$, the optimal value of (s,t,θ) is (1.1 mm, 3 mm, 20°), which leading to approximately 4% increase of the spatially-averaged adiabatic film cooling effectiveness related to the reference case. Under a high blowing ratio of $M = 1.5$, the optimal value of (s,t,θ) is (2.2 mm, 3.9 mm, 29.1°), which leading to approximately 23% increase related to the reference case.

The optimal round-to-slot hole produces more reasonable mutual interaction between the coolant jet and the primary flow, resulting an improvement in the lateral spreading of the coolant jet downstream the film-hole exit as well as the effective enhancement of local adiabatic film cooling effectiveness in the region apart from the film-hole centerline, especially under a high blowing ratio. In general, the optimal round-to-slot hole is of converging feature under a low blowing ratio. However, under a high blowing ratio, the optimal round-to-slot hole is of diffusing feature where the slot area at the film-hole exit is lager than the round cross section at the film-hole inlet.

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